

Lab 2: The Linear Model (Data 622)

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```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import statsmodels.formula.api as smf
import statsmodels.api as sm
from statsmodels.stats.outliers_influence import variance_inflation_factor

sns.set_style("whitegrid")
pd.set_option('display.max_columns', 20)

df = pd.read_csv("earnings.csv")
print(f"Dataset shape: {df.shape}")
df.head()
```

Dataset shape: (1816, 15)

	height	weight	male	earn	earnk	ethnicity	education	mother_education	father_education
0	74	210.0	1	50000.0	50.0	White	16.0	12.0	12.0
1	66	125.0	0	60000.0	60.0	White	16.0	12.0	12.0
2	64	126.0	0	30000.0	30.0	White	16.0	12.0	12.0
3	65	200.0	0	25000.0	25.0	White	17.0	12.0	NaN
4	63	110.0	0	50000.0	50.0	Other	16.0	15.0	12.0

Problem 1: Confounds in Earnings Data

(a) Exploratory Data Analysis

Inspect the data structure and missing values

Column dtypes:

```
height          int64
weight          float64
male            int64
earn            float64
earnk           float64
ethnicity       str
education       float64
mother_education float64
father_education float64
walk            int64
exercise        int64
smokenow       float64
tense          float64
angry           int64
age            int64
dtype: object
```

Basic statistics:

	height	weight	male	earn	earnk	education	mother_education	father_education
count	1816.00	1789.00	1816.00	1816.00	1816.00	1814.00	1572.00	1521.00
mean	66.57	156.31	0.37	21147.30	21.15	13.24	11.19	11.22
std	3.83	34.62	0.48	22531.77	22.53	2.56	3.15	3.73
min	57.00	80.00	0.00	0.00	0.00	2.00	0.00	0.00
25%	64.00	130.00	0.00	6000.00	6.00	12.00	9.00	8.00
50%	66.00	150.00	0.00	16000.00	16.00	12.00	12.00	12.00
75%	69.25	180.00	1.00	27000.00	27.00	15.00	12.00	13.00
max	82.00	342.00	1.00	400000.00	400.00	18.00	18.00	18.00

Missing values per column:

	count	pct
father_education	295	16.24
mother_education	244	13.44
weight	27	1.49
education	2	0.11
tense	2	0.11
smokenow	1	0.06

Handling Missing Values:

The dataset has 1816 observations. The missingness pattern is as follows: `father_education` is missing for 16.2% of rows, `mother_education` for 13.4%, `weight` for 1.5%, and `education`, `tense`, and `smokenow` each have 1-2 missing values. No rows have missing `earnk`, `height`, `male`, `age`, `walk`, `exercise`, `angry`, or `ethnicity`.

My strategy:

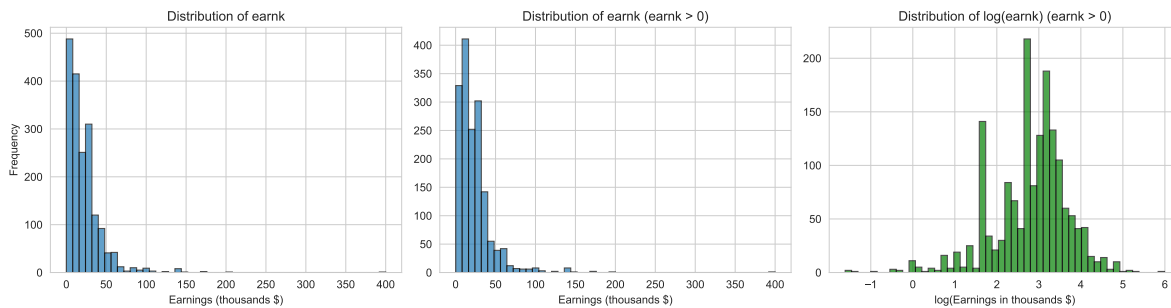
- **earnk (target)**: No values are missing, but 187 observations (10.3%) have zero earnings. These represent unemployed individuals, stay-at-home parents, or other non-earners. I retain them in the dataset but address them in the log transform discussion below.
- **weight** (27 missing): Per the data dictionary, these are cases where weight exceeded 500 lbs or was not recorded. I retain these rows and allow listwise deletion in models that use `weight`.
- **mother_education and father_education** (244 and 295 missing): These represent “Don’t Know” or blank survey responses. Because missingness is substantial (~15%) and likely not random (respondents who don’t know their parents’ education may differ systematically from those who do), I do not impute. I let listwise deletion handle these in relevant models, and I note that this reduces the sample size.
- **education, tense, smokenow** (1-2 missing each): Negligible missingness; listwise deletion has minimal impact.

Total rows: 1816

Rows with `earnk == 0`: 187 (10.3%)

Rows with `earnk > 0`: 1629

Distribution of `earnk` (target variable)



Discussion: Log Transform of `earnk`

The raw distribution of `earnk` is strongly right-skewed: the median is \$16k, the mean is \$21k, and the maximum is \$400k. A small number of high earners create a long right tail. This skewness causes two problems for OLS regression: the residuals will be non-normal, and a few extreme observations will have outsized influence on the fitted line (high leverage).

Pros of log transform:

The log-transformed distribution (restricted to positive earners) is much more symmetric and approximately normal, better satisfying the homoscedasticity and normality assumptions of OLS. On a log scale, coefficients have a natural *multiplicative* (percentage) interpretation: a coefficient of 0.05 on a predictor means a one-unit increase in that predictor is associated with approximately a 5% increase in earnings. This is also more natural for income data, since the economic “meaning” of a \$5,000 raise differs greatly depending on whether your baseline is \$10k or \$100k. The log transform also reduces the influence of high-income outliers.

Cons of log transform:

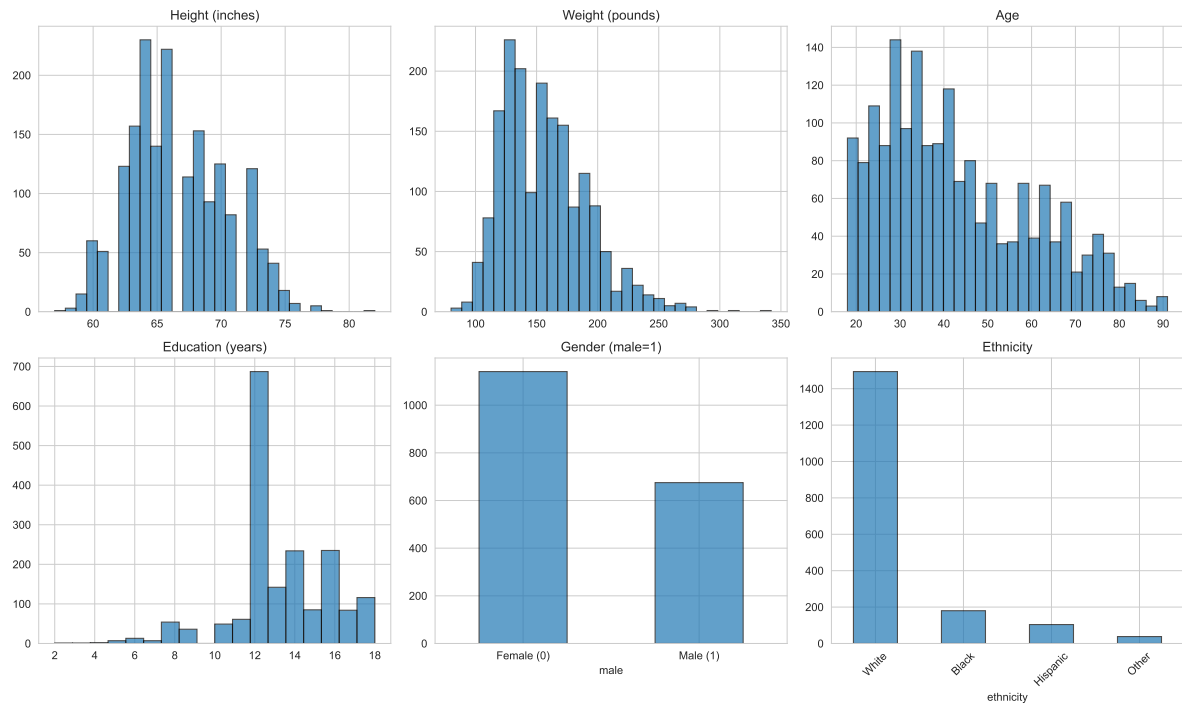
We must exclude 187 observations (10.3%) with zero earnings. These zero earners are a substantively important group — they include the unemployed, retirees, stay-at-home parents, and others. Dropping them introduces **selection bias**: we are effectively modeling earnings *conditional on having positive income*, which is a different question than modeling earnings for the full population. Factors like gender and education affect both *whether* someone earns and *how much* they earn, so restricting to positive earners can distort estimated effects. Additionally, the percentage-change interpretation may be less intuitive for some audiences.

Decision: I use `log(earnk)` as my target, restricting to respondents with positive earnings. This is standard in labor economics (Mincer earnings equations). The log-normal model better satisfies OLS assumptions. However, I acknowledge that excluding zero earners introduces selection bias, and findings should be interpreted as effects on earnings *among those who earn*.

Rows with positive earnings (used in modeling): 1629

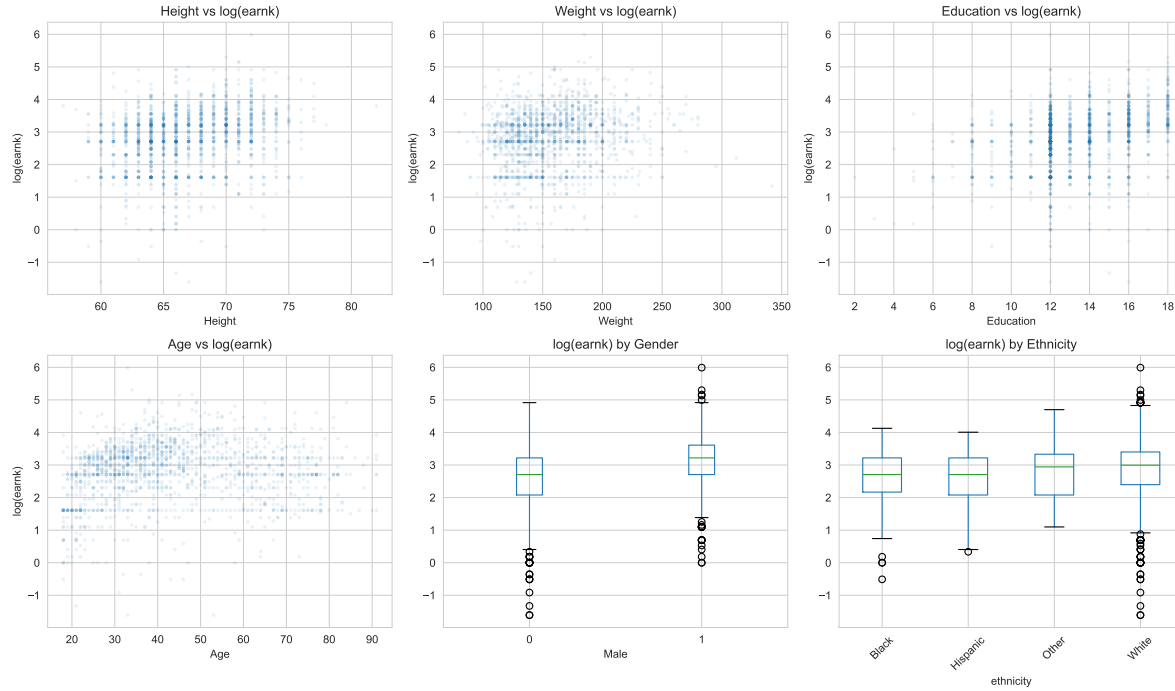
Rows excluded (`earnk = 0`): 187

Key variable distributions

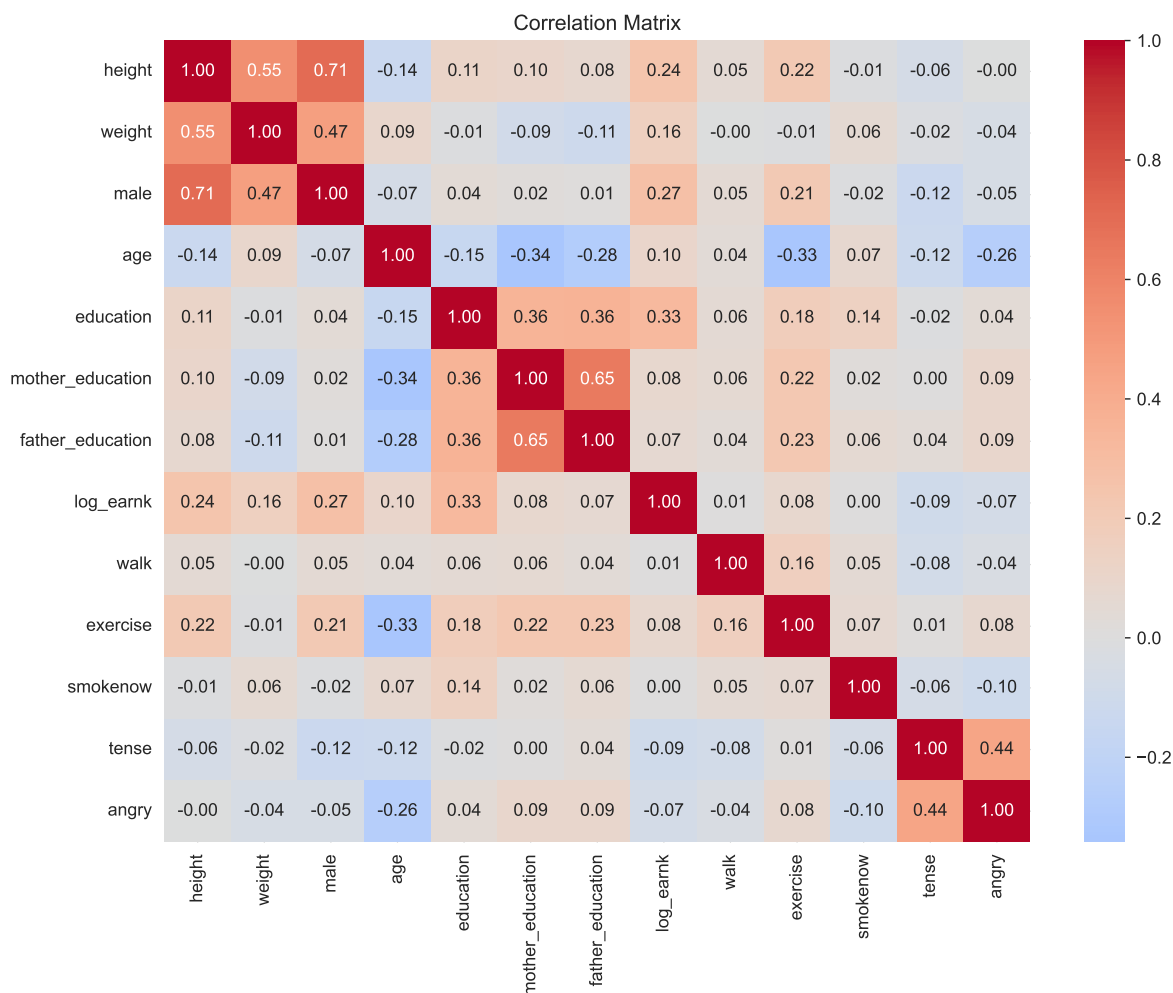


Notable observations from the distributions: the sample is approximately 63% female and 37% male. The education distribution is concentrated around 12 (high school) and 16 (college) years. The sample is predominantly White, with smaller Black, Hispanic, and Other groups. Age ranges from about 18 to 90 with a fairly uniform distribution.

Key relationships with earnings



The scatter plots reveal several patterns. Education has the clearest positive linear relationship with log earnings. Height shows a moderate positive association, but this is likely confounded by gender (as explored in part b). Age shows a hump-shaped pattern earnings rise through middle age and decline later suggesting a quadratic term would be appropriate. The boxplots confirm that males have higher median log earnings, and there are differences across ethnic groups, with White respondents having the highest median earnings in this 1990 sample.



The correlation matrix reveals key relationships relevant to confounding. The strongest correlation is between `height` and `male` ($r = 0.71$), followed by `mother_education` and `father_education` ($r = 0.65$), and `height` and `weight` ($r = 0.55$). These high correlations foreshadow the multicollinearity and confounding issues explored below. The correlation of `log_earnk` with `education` (0.33) and `male` (0.27) are the strongest predictor-target correlations.

(b) Height and Weight Confounds

Model 1: Height only

OLS Regression Results

=====

```

Dep. Variable:          log_earnk    R-squared:          0.060
Model:                  OLS          Adj. R-squared:      0.059
Method:                 Least Squares  F-statistic:        103.0
Date:                   Sat, 14 Feb 2026  Prob (F-statistic):  1.65e-
23
Time:                   14:35:20      Log-Likelihood:      -
2097.1
No. Observations:       1629         AIC:                4198.
Df Residuals:           1627         BIC:                4209.
Df Model:                1
Covariance Type:        nonrobust

```

```

=====
              coef      std err          t      P>|t|      [0.025      0.975]
-----
Intercept    -0.9991      0.376      -2.657      0.008      -1.737      -
0.261
height        0.0570      0.006      10.149      0.000       0.046      0.068
=====
Omnibus:                209.188    Durbin-Watson:           1.878
Prob(Omnibus):           0.000    Jarque-Bera (JB):        375.029
Skew:                    -0.827    Prob(JB):                3.66e-
82
Kurtosis:                4.671    Cond. No.                1.16e+03
=====

```

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The condition number is large, 1.16e+03. This might indicate that there are strong multicollinearity or other numerical problems.

Height coefficient 95% CI:

0 0.046016

1 0.068062

Name: height, dtype: float64

Model 2: Weight only

```

              OLS Regression Results
=====
Dep. Variable:          log_earnk    R-squared:          0.024
Model:                  OLS          Adj. R-squared:      0.024
Method:                 Least Squares  F-statistic:        39.86

```



```

Date:                Sat, 14 Feb 2026    Prob (F-statistic):        3.53e-
10
Time:                14:35:20    Log-Likelihood:            -
2099.7
No. Observations:    1607    AIC:                4203.
Df Residuals:        1605    BIC:                4214.
Df Model:            1
Covariance Type:      nonrobust
=====
               coef      std err          t      P>|t|      [0.025      0.975]
-----
Intercept        2.1723      0.104     20.856     0.000      1.968      2.377
weight           0.0041      0.001      6.313     0.000      0.003      0.005
=====
Omnibus:                193.642    Durbin-Watson:           1.864
Prob(Omnibus):           0.000    Jarque-Bera (JB):        338.113
Skew:                   -0.793    Prob(JB):                3.80e-
74
Kurtosis:                4.591    Cond. No.                753.
=====

```

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Weight coefficient 95% CI:

0 0.002812

1 0.005347

Name: weight, dtype: float64

Model 3: Height and Weight together

```

                                OLS Regression Results
=====
Dep. Variable:                log_earnk    R-squared:                0.060
Model:                        OLS          Adj. R-squared:          0.059
Method:                        Least Squares    F-statistic:            51.00
Date:                Sat, 14 Feb 2026    Prob (F-statistic):        3.37e-
22
Time:                14:35:20    Log-Likelihood:            -
2069.9
No. Observations:    1607    AIC:                4146.
Df Residuals:        1604    BIC:                4162.

```

```

Df Model:                2
Covariance Type:        nonrobust
=====
              coef      std err          t      P>|t|      [0.025      0.975]
-----
Intercept    -0.8702      0.404     -2.155     0.031     -1.662         -
0.078
height        0.0536      0.007      7.788     0.000      0.040      0.067
weight        0.0007      0.001      0.855     0.393     -0.001      0.002
=====
Omnibus:                207.303   Durbin-Watson:                1.873
Prob(Omnibus):           0.000   Jarque-Bera (JB):           373.347
Skew:                    -0.828   Prob(JB):                   8.49e-
82
Kurtosis:                4.684   Cond. No.                   3.21e+03
=====

```

Notes:

- [1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
- [2] The condition number is large, 3.21e+03. This might indicate that there are strong multicollinearity or other numerical problems.

Coefficient 95% CIs:

```

              0              1
Intercept -1.662232 -0.078078
height     0.040092  0.067084
weight     -0.000854  0.002173

```

What happened when height and weight are combined?

When fit separately, both height and weight are statistically significant positive predictors of log earnings. Height has a coefficient of approximately 0.057 (95% CI: [0.046, 0.068], $p < 0.001$), meaning each additional inch of height is associated with roughly 5.7% higher earnings. Weight has a smaller but still significant coefficient of approximately 0.004 (95% CI: [0.003, 0.005], $p < 0.001$).

However, when both variables are included together, the weight coefficient drops dramatically to approximately 0.0007 and becomes *non-significant* ($p \approx 0.39$), while the height coefficient barely changes (≈ 0.054). This occurs because height and weight are correlated ($r = 0.55$): taller people tend to weigh more. Once we account for height, weight adds almost no additional explanatory power. The apparent bivariate relationship between weight and earnings was largely driven by the shared variation with height — weight was acting as a noisy proxy for height, not as an independent predictor.

But why does height predict earnings? The key confound is **gender**.

Model 4: Adding gender as a confounder

OLS Regression Results						
Dep. Variable:	log_earnk	R-squared:	0.080			
Model:	OLS	Adj. R-squared:	0.078			
Method:	Least Squares	F-statistic:	46.35			
Date:	Sat, 14 Feb 2026	Prob (F-statistic):	1.01e-			
28						
Time:	14:35:20	Log-Likelihood:	-			
2052.6						
No. Observations:	1607	AIC:	4113.			
Df Residuals:	1603	BIC:	4135.			
Df Model:	3					
Covariance Type:	nonrobust					
	coef	std err	t	P> t	[0.025	0.975]
Intercept	1.0879	0.519	2.095	0.036	0.070	2.106
height	0.0236	0.008	2.780	0.006	0.007	0.040
weight	-2.366e-05	0.001	-0.031	0.976	-0.002	0.001
male	0.3723	0.063	5.907	0.000	0.249	0.496
Omnibus:	211.206	Durbin-Watson:	1.861			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	380.424			
Skew:	-0.841	Prob(JB):	2.47e-			
83						
Kurtosis:	4.689	Cond. No.	4.18e+03			

Notes:

- [1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
 [2] The condition number is large, 4.18e+03. This might indicate that there are strong multicollinearity or other numerical problems.

Coefficient 95% CIs:

	0	1
Intercept	0.069500	2.106272
height	0.006951	0.040266
weight	-0.001538	0.001491

male 0.248672 0.495905

Interpretation with gender included:

After adding `male` to the model, the height coefficient drops by more than half from 0.054 to approximately 0.024 and the weight coefficient is essentially zero (≈ -0.00002 , $p \approx 0.98$). Meanwhile, `male` has a large and highly significant coefficient of approximately 0.37 ($p < 0.001$).

This is textbook confounding. Gender (`male`) is correlated with both the predictors (men are taller, $r = 0.71$, and heavier, $r = 0.47$) and the outcome (men earn more). The causal structure is:

- Male $\rightarrow$ taller/heavier (biological)
- Male $\rightarrow$ higher earnings (labor market gender gap in 1990)
- Therefore: taller/heavier $\rightarrow$ higher earnings (spurious, driven by gender)

Once we control for gender, most of the apparent height “premium” evaporates. The residual height effect (coef ≈ 0.024 , still significant at $p = 0.006$) may reflect a genuine but small height premium documented in the labor economics literature, or it could be explained by additional confounders. The weight effect disappears entirely.

This demonstrates why including confounders is essential: without controlling for gender, we would falsely conclude that gaining weight increases your earnings.

(c) Education Coefficients

Model 1: Parental education only (total effect)

Model: Parental Education Only

OLS Regression Results

```
=====
Dep. Variable:          log_earnk      R-squared:                0.006
Model:                  OLS            Adj. R-squared:           0.004
Method:                 Least Squares   F-statistic:              3.734
Date:                  Sat, 14 Feb 2026 Prob (F-statistic):      0.0242
Time:                  14:35:20         Log-Likelihood:           -
1750.1
No. Observations:      1315            AIC:                    3506.
Df Residuals:          1312            BIC:                    3522.
Df Model:               2
Covariance Type:       nonrobust
=====
```

	coef	std err	t	P> t	[0.025	0.975]
Intercept	2.5996	0.098	26.543	0.000	2.407	2.792
mother_education	0.0142	0.011	1.343	0.179	-0.007	0.035
father_education	0.0085	0.009	0.958	0.338	-0.009	0.026
=====						
Omnibus:		195.665	Durbin-Watson:		1.840	
Prob(Omnibus):		0.000	Jarque-Bera (JB):		392.711	
Skew:		-0.886	Prob(JB):		5.30e-	
86						
Kurtosis:		5.007	Cond. No.		64.5	
=====						

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Model 2: Parental education + respondent's education

Model: Parental Education + Own Education

OLS Regression Results

```

=====
Dep. Variable:          log_earnk      R-squared:                0.098
Model:                  OLS           Adj. R-squared:           0.096
Method:                 Least Squares  F-statistic:              47.58
Date:                  Sat, 14 Feb 2026 Prob (F-statistic):       3.46e-
29
Time:                  14:35:20        Log-Likelihood:          -
1684.7
No. Observations:      1314           AIC:                    3377.
Df Residuals:          1310           BIC:                    3398.
Df Model:               3
Covariance Type:       nonrobust
=====

```

	coef	std err	t	P> t	[0.025	0.975]
Intercept	1.2571	0.149	8.457	0.000	0.965	1.549
mother_education	-0.0041	0.010	-0.403	0.687	-0.024	0.016
father_education	-0.0100	0.009	-1.170	0.242	-0.027	0.007
education	0.1277	0.011	11.597	0.000	0.106	0.149
=====						
Omnibus:		243.221	Durbin-Watson:		1.880	
Prob(Omnibus):		0.000	Jarque-Bera (JB):		587.267	

Skew:	-1.004	Prob(JB):	3.00e-
128			
Kurtosis:	5.587	Cond. No.	133.
=====			

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Interpretation:

In the first model (parental education only), the coefficients capture the **total effect** of parental education on earnings. `mother_education` has a coefficient of approximately 0.014 and `father_education` approximately 0.009. These are relatively small and individually non-significant ($p = 0.18$ and $p = 0.34$ respectively), though the joint F-test of the model is significant ($p = 0.024$). The total effect of parental education on earnings is modest — a combined one-year increase in both parents’ education is associated with roughly a 2.3% increase in earnings.

In the second model (adding the respondent’s own `education`), something striking happens: the parental education coefficients **reverse sign**. `mother_education` becomes -0.004 and `father_education` becomes -0.010 — both are now negative and non-significant. Meanwhile, `education` has a large coefficient of approximately 0.128 ($p < 0.001$), and the model R^2 jumps from 0.006 to 0.098.

This sign reversal is explained by the mediator structure. The respondent’s own education is a **mediator** on the causal path: Parental Education $\rightarrow$ Respondent’s Education $\rightarrow$ Earnings. Parents with more education raise children who attain more education ($r \approx 0.36$ between parental and own education), and more education leads to higher earnings.

In the first model, the parental education coefficients absorb the *total* effect both the direct effect of parental education (social capital, networks, cultural knowledge) and the indirect effect flowing through the child’s own education. In the second model, by conditioning on own education, we block the indirect path. The parental coefficients now represent only the *direct* effect of parental education on earnings, holding the child’s education constant. The fact that these direct effects are essentially zero (and even slightly negative) suggests that virtually all of the parental education effect on earnings operates *indirectly* through its influence on the child’s own educational attainment.

The slight negative coefficients (though not significant) could arise from a form of Simpson’s paradox: among people with the *same* level of education, those whose parents had higher education may have “underperformed” relative to their upbringing, while those from less-educated families who achieved the same education level may be more driven or talented — a selection effect. However, these coefficients are well within the range of sampling noise, and we should not over-interpret non-significant results.

(d) Potential Colliders

Collider bias arises when we condition on a variable that is a common *effect* of two other variables. Conditioning on such a variable (by including it in a regression or by sample restriction) can create a spurious association between its causes, even if they are independent in the population.

I identify the following potential collider concerns in this dataset:

1. Sample restriction to positive earners ($\text{earnk} > 0$): This is arguably the most important collider concern. By taking the log transform, we restrict the sample to people who earn positive income. Labor force participation (having positive earnings) is itself influenced by both gender and education — women were less likely to work full-time in 1990, and more educated people are more likely to work. Conditioning on having positive earnings is equivalent to conditioning on a collider on the path:

$$\text{Gender} \rightarrow \text{Positive Earnings} \leftarrow \text{Education}$$

This means that among positive earners only, gender and education may appear to have a relationship that differs from the full population. For instance, among women who *do* earn positive income, education may have a stronger effect on earnings because only the most educated or motivated women entered the labor force in 1990. This is a form of **selection bias** related to collider conditioning.

2. Health behavior variables (*walk*, *exercise*, *smokenow*, *tense*, *angry*): These are plausibly caused by multiple other variables in the model. For instance, *exercise* frequency is influenced by both education (more educated people exercise more, $r = 0.18$) and by earnings (higher income provides more leisure time and gym access). Similarly, *tense* and *angry* may be affected by both work conditions (related to earnings) and by demographic factors. If exercise or stress are colliders on certain causal paths, including them could open backdoor paths and distort the estimated effects of education or gender on earnings. For example, if both high education and high earnings independently cause more exercise, then conditioning on exercise would induce a spurious *negative* association between education and earnings through the opened collider path.

3. Weight as a potential collider: Weight is influenced by socioeconomic status (through diet quality, stress, healthcare access — all affected by income and education) and also by health behaviors like exercise and age. If weight sits at the intersection of multiple causal pathways, conditioning on it in a model that includes SES and health variables could distort the relationships among these predictors.

4. Overcontrol bias via education: While not a classical collider, conditioning on the respondent’s own education when estimating the effect of parental education (as in part c) blocks the primary causal pathway and biases the estimate of the total effect toward zero. As we saw, the coefficients actually reversed sign. This is sometimes called “overcontrol bias” or

“mediator bias” — it is structurally related to collider bias in that conditioning on a descendant of the treatment distorts the causal estimate.

The lesson is that variable inclusion decisions must be guided by a **directed acyclic graph (DAG)** representing the assumed causal structure. Blindly including all available variables or selecting variables based solely on statistical significance and model fit can introduce as much bias as it removes. For each regression, we should ask: “What causal effect am I trying to estimate?” and include only the variables that block backdoor paths without opening collider paths or blocking the causal path of interest.

Problem 2: Optimizing a Predictive Model

Important methodological note: AIC values can only be meaningfully compared across models fit on *the same set of observations* with the *same target variable*. Because several variables have missing values (especially `mother_education` and `father_education`), different model specifications would use different sample sizes due to listwise deletion, making their AIC values incomparable. To address this, I create a **common complete-case subset** that includes all variables I might use, and fit every candidate model on this same subset.

Complete-case subset size: 1295 (out of 1816 total, out of 1629 positive earners)
 Rows excluded due to missing values in any variable: 334

(a) Baseline Model

Baseline Model: age + male + education

OLS Regression Results					
=====					
Dep. Variable:	log_earnk	R-squared:	0.212		
Model:	OLS	Adj. R-squared:	0.210		
Method:	Least Squares	F-statistic:	115.6		
Date:	Sat, 14 Feb 2026	Prob (F-statistic):	2.75e-		
66					
Time:	14:35:20	Log-Likelihood:	-		
1576.0					
No. Observations:	1295	AIC:	3160.		
Df Residuals:	1291	BIC:	3181.		
Df Model:	3				
Covariance Type:	nonrobust				
=====					
	coef	std err	t	P> t	[0.025 0.975]

Intercept	0.4649	0.149	3.122	0.002	0.173	0.757
age	0.0122	0.001	8.619	0.000	0.009	0.015
male	0.5147	0.046	11.117	0.000	0.424	0.606
education	0.1222	0.010	12.748	0.000	0.103	0.141

Omnibus:	250.776	Durbin-Watson:	1.898
Prob(Omnibus):	0.000	Jarque-Bera (JB):	601.044
Skew:	-1.050	Prob(JB):	3.05e-
131			
Kurtosis:	5.595	Cond. No.	301.

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Deviance (sum of squared residuals): 864.6722

AIC: 3159.9803

The baseline model with `age`, `male`, and `education` explains approximately 20% of the variance in log earnings ($R^2 \approx 0.21$). All three predictors are highly significant. Each additional year of education is associated with roughly 12% higher earnings. Being male is associated with roughly 65% higher earnings ($\exp(0.51) \approx 1.66$). Each additional year of age is associated with about 1% higher earnings, though this linear specification misses the clearly non-linear (hump-shaped) age-earnings profile visible in the EDA.

(b) Feature Engineering and Variable Selection

Feature engineering

Working dataframe shape: (1295, 24)

Engineered features:

<code>age_sq:</code>	<code>age^2</code> (captures hump-shaped age-earnings profile)
<code>age_male:</code>	<code>age x male</code> (gender-specific age trajectories)
<code>education_sq:</code>	<code>education^2</code> (non-linear returns to schooling)
<code>parent_education_avg:</code>	mean of mother and father education
<code>education_male:</code>	<code>education x male</code> (gender-specific education returns)
<code>activity_index:</code>	mean of walk and exercise frequency
<code>stress_index:</code>	mean of tense and angry days
<code>bmi:</code>	body mass index from height and weight

Feature engineering rationale:

The most important engineered feature is **age_sq**. The EDA scatter plot of age vs. log earnings clearly shows a hump-shaped pattern: earnings rise steeply in early career, plateau around age 40-50, and decline among older workers (retirement, reduced hours). A linear age term forces a monotonic relationship and cannot capture this curvature. Including **age_sq** allows the model to fit a parabola, which should substantially improve fit. This is the standard specification in Mincer earnings equations in labor economics.

The **education_male** interaction captures the possibility that the earnings return per year of education differs between men and women. In 1990, occupational segregation and discrimination may have meant that the earnings premium for additional education was not identical across genders.

I combine **mother_education** and **father_education** into a single average (**parent_education_avg**) because they are highly correlated ($r = 0.65$). Including both separately creates multicollinearity without much additional predictive power compared to their average, and using the average provides a more stable single-parameter estimate.

Model comparisons

Model	AIC	BIC	R-sq	Adj R-sq	N
M10: + bmi	3048.74	3100.40	0.2833	0.2782	1295
M3: + parent_ed	3050.86	3081.86	0.2776	0.2748	1295
M9: simple+parent	3050.86	3081.86	0.2776	0.2748	1295
M4: + ethnicity	3054.93	3101.43	0.2787	0.2742	1295
M11: + stress	3055.13	3106.80	0.2797	0.2747	1295
M5: + interactions	3055.42	3112.25	0.2807	0.2751	1295
M12: tuned	3055.63	3107.29	0.2794	0.2744	1295
M6: + health	3058.03	3130.36	0.2825	0.2753	1295
M2: + age_sq	3058.77	3084.60	0.2721	0.2698	1295
M7: + ed_sq	3059.03	3131.36	0.2820	0.2747	1295
M8: parsimonious	3062.54	3109.04	0.2745	0.2699	1295
M1: baseline	3159.98	3180.65	0.2117	0.2099	1295

Unique sample sizes across all models: {1295}

All models use the same observations - AIC values are directly comparable.

Best model summary

Best model by AIC: M10: + bmi
AIC: 3048.74

OLS Regression Results						
=====						
Dep. Variable:	log_earnk	R-squared:	0.283			
Model:	OLS	Adj. R-squared:	0.278			
Method:	Least Squares	F-statistic:	56.42			
Date:	Sat, 14 Feb 2026	Prob (F-statistic):	8.47e-			
87						
Time:	14:35:20	Log-Likelihood:	-			
1514.4						
No. Observations:	1295	AIC:	3049.			
Df Residuals:	1285	BIC:	3100.			
Df Model:	9					
Covariance Type:	nonrobust					
=====						
	coef	std err	t	P> t	[0.025	0.975]

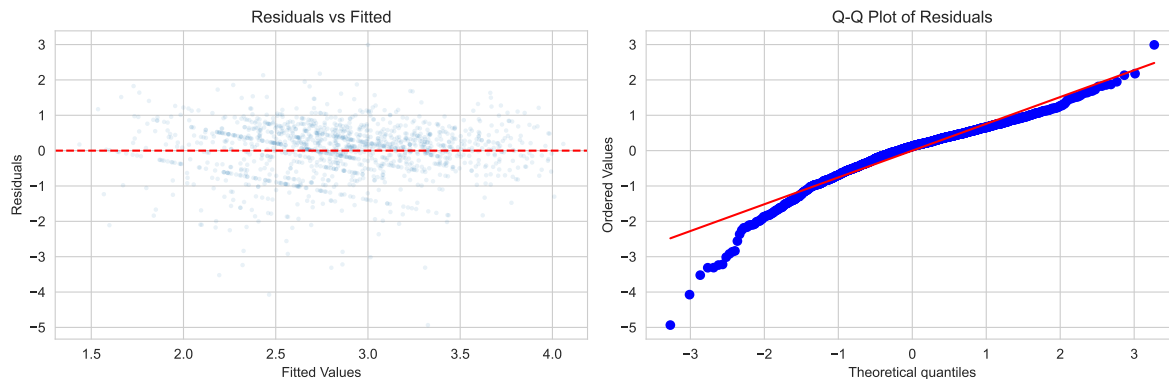
Intercept	-0.8329	0.242	-3.443	0.001	-	
1.307 -0.358						
C(ethnicity) [T.Hispanic]	0.0428	0.118	0.362	0.718	-	
0.190 0.275						
C(ethnicity) [T.Other]	0.1537	0.171	0.900	0.368	-	
0.181 0.489						
C(ethnicity) [T.White]	0.0841	0.077	1.088	0.277	-	
0.068 0.236						
age	0.0961	0.008	12.553	0.000	0.081	0.111
age_sq	-0.0009	7.79e-05	-11.081	0.000	-	
0.001 -0.001						
male	0.5470	0.045	12.051	0.000	0.458	0.636
education	0.0910	0.010	8.888	0.000	0.071	0.111
parent_education_avg	0.0226	0.008	2.709	0.007	0.006	0.039
bmi	-0.0157	0.005	-2.856	0.004	-	
0.026 -0.005						
=====						
Omnibus:	276.980	Durbin-Watson:	1.939			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	841.076			
Skew:	-1.065	Prob(JB):	2.30e-			
183						

Kurtosis:	6.324	Cond. No.	2.85e+04
=====			

Notes:

- [1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
- [2] The condition number is large, 2.85e+04. This might indicate that there are strong multicollinearity or other numerical problems.

Residual diagnostics for the best model



Discussion of the best model and variable selection:

Across all 12 candidate models (fit on the same complete-case subset for valid AIC comparison), the key findings are:

Most impactful improvement `age_sq`: Adding the quadratic age term is by far the single biggest improvement over the baseline. Moving from M1 (linear age only) to M2 (with `age_sq`) produces the largest AIC drop in the entire model search. This makes sense: the age-earnings relationship is clearly non-linear, and a linear specification severely misrepresents it. The negative `age_sq` coefficient confirms the inverted-U shape: earnings rise through prime working years and decline afterwards.

Parental education (`parent_education_avg`): Adding average parental education provides a consistent AIC improvement beyond education alone. Even after controlling for the respondent's own education, parental education has a small positive coefficient. In a *predictive* context (unlike the inference question in 1c), this is useful because it captures additional signal — perhaps reflecting social capital, financial support, or neighborhood effects that are correlated with but not fully explained by own education.

Ethnicity: Adding ethnicity dummies provides a further small AIC improvement, reflecting earnings disparities across racial/ethnic groups that persist after controlling for education and other variables.

Interactions (`education_male`, `age_male`): The `education_male` interaction is my candidate for most useful engineered feature. It allows the model to estimate different returns to education for men vs. women, which is economically plausible given the labor market conditions of 1990. However, the AIC improvement from interactions is modest and not always clearly better than simpler models — the extra parameters are penalized by AIC.

Excluded variables and reasoning:

- **height and weight** as raw variables: Excluded because their effects are largely confounded by gender (as shown in 1b). Including them alongside `male` provides no meaningful AIC improvement. However, `bmi` which combines height and weight into a single health-related metric does appear in the best model (M10). Unlike raw height, BMI captures body composition rather than stature, and its significant negative coefficient (-0.016 , $p = 0.004$) suggests that higher BMI is associated with lower earnings even after controlling for gender, possibly reflecting health-related productivity effects or workplace discrimination based on body weight.
- **mother_education and father_education individually:** Not included separately because they are highly collinear ($r = 0.65$). Their average `parent_education_avg` captures equivalent information with half the parameters.
- **Health variables (`walk`, `exercise`, `smokenow`, `tense`, `angry`):** Including these individually or as indices (`activity_index`, `stress_index`) does not improve AIC. They have weak marginal associations with earnings after controlling for demographic and education variables. Additionally, as noted in Problem 1(d), some may act as colliders.
- **education_sq:** Does not improve the model, suggesting returns to education are approximately log-linear (each additional year provides roughly the same percentage increase in earnings).

Collinear pairs avoided: I do not include both `height` and `weight` together ($r = 0.55$), nor both `mother_education` and `father_education` separately ($r = 0.65$). In each case I use the more parsimonious alternative (`height` alone or `parent_education_avg`). Including collinear pairs inflates standard errors, wastes degrees of freedom, and does not improve predictive performance.